Mark A. Nicholas

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EXPERIENCE AND EDUCATION

Ph.D., Biological Sciences - Carnegie Mellon University 2020 - 2025

Thesis: Corticalstriatal function during movement and trial and error learning in a dynamic environment

Advisor: Dr. Eric A. Yttri, Department of Biological Sciences

Center for the Neural Basis of Cognition

Research Associate – Eric Yttri Lab, Carnegie Mellon University 2017 - 2020

Promoted from Research Associate I to Research Associate II in 2019

Postbaccalaureate Intramural Research Training Fellow - NIMH 2015 - 2017

Laboratory of Neuropsychology, Section on Neurobiology and Learning and Memory

Advisor: Dr. Elisabeth A. Murray

Research Assistant – Sandra Kuhlman Lab, Carnegie Mellon University 2012 - 2015

Project: Corticocortical feedback-mediated facilitation is cell-type specific and gated by nucleus basalis activity

Bachelor of Science, Neuroscience - Carnegie Mellon University 2011 - 2015

College Honors, Department of Biological Sciences

AWARDS AND HONORS

Probing Sensorimotor Circuits and Perception Across Species with Microstimulation People's Choice Poster Award 2025

Center for the Neural Basis of Cognition McClelland Award for Outstanding Student Paper 2025

Center for the Neural Basis of Cognition Special Interest Group Award 2025

Biological Sciences Stupakoff Outstanding Research Paper Award 2025

Biological Sciences Margaret Carver Travel Grant 2025

Mellon College of Science Graduate Student Conference Funding Award 2024

Biological Sciences Margaret Carver Travel Grant 2023

Mellon College of Science Rookie of the Year Finalist 2018, 2019

One of 6 employees nominated from the entire college

National Institutes of Health Postbaccalaureate Intramural Research Training Award 2015, 2016

Mellon College of Science College Honors, Carnegie Mellon University 2015

Senior Leadership Recognition, Carnegie Mellon University 2015

"This recognition is reserved for those students who have made an unparalleled impact on our community, leaving CMU a better place as a result of their leadership, vision and initiative"

Small Undergraduate Research Grant, Carnegie Mellon University 2015

Howard Hughes Medical Institute Summer Researcher Grant 2013, 2014

PUBLICATIONS

1. **Nicholas MA**, Yttri EA. (2024). *Motor cortex is responsible for motoric dynamics in striatum and the execution of both skilled and unskilled actions.* Neuron.

2. Kaskan PM*, **Nicholas MA***, Dean, AM, Murray, EM. (2022). *Attention to stimuli of learned versus innate biological value rely on separate neural systems.* Journal of Neuroscience (* contributed equally)

3. Saleh MS*, Ritchie SM*, **Nicholas MA***, Gordon HL*, Yuan B, Hu C, Jahan S, Bezbaruah R, Reddy JW, Chamanzar M, Yttri EA, Panat RP. (2022). *CMU Array: A 3D nanoprinted, fully customizable high-density microelectrode array platform.* Science Advances. (* contributed equally)

4. Belsey P, **Nicholas MA**, Yttri EA. (2020). *Open-source joystick manipulandum for decision-making, reaching, and motor control studies in mice.* eNeuro.

5. Saleh MS, Ritchie S, **Nicholas MA**, Bezbaruah R, Panat R, Yttri EA. (2019). *CMU Array: A 3D Nano-Printed, Customizable Ultra-High-Density Microelectrode Array Platform.* bioRxiv.

6. Belsey P, **Nicholas MA**, Yttri EA. (2019). *Build a better mouse task: can an open-source rodent joystick enhance reaching behavior outcomes through improved monitoring of real-time spatiotemporal kinematics?* bioRxiv.

7. Kaskan PM, Dean MA, **Nicholas MA**, Mitz AR, Murray EA. (2018). *Gustatory responses in macaque monkeys revealed with fMRI: comments on taste, taste preference and internal state.* NeuroImage.

8. Pafundo DE, **Nicholas MA**, Zhang R, Kuhlman SJ. (2016). *Top-Down-Mediated Facilitation in the Visual Cortex Is Gated by Subcortical Neuromodulation.* Journal of Neuroscience.

9. **Nicholas MA**, Pafundo DE. (2014). *Can we reorganize the cortex to restore vision in amblyopes?* The Triple Helix, The Journal of Science, Society, and Law at Carnegie Mellon.

PUBLICATIONS IN PREPARATION

1. **Nicholas MA**, Yttri EA. *Corticostriatal function during skilled motor actions and trial and error learning in a dynamic environment.*

2. Hsu AI, **Nicholas MA**, Yttri EA. *Naturalistic action encoding across corticostriatal motor axis.*

PRESENTATIONS

1. **Nicholas MA**. (2025). *Motor cortex is responsible for motoric dynamics in striatum and the execution of both skilled and unskilled actions.* ALM / Cortical Circuits, International Online Journal Club, invited talk.

2. **Nicholas MA**, Yttri EA. (2025). *"Corticostriatal contributions to skilled motor actions and trial and error learning.* Probing Sensorimotor Circuits and Perception Across Species with Microstimulation, poster.

3. **Nicholas MA**, Yttri EA. (2025). *"Corticostriatal contributions to skilled motor actions and trial and error learning.* Society for the Neural Control of Movement, poster.

4. **Nicholas MA**. (2025). *Motor cortex is responsible for motoric dynamics in striatum and the execution of both skilled and unskilled actions.* Systems, Behavioral, and Computational Neuroscience Journal Club. Georgia Institute of Technology, invited talk.

5. **Nicholas MA**, Yttri EA. (2024). *Cell-type-specific corticostriatal dynamics orchestrate trial and error motor learning.* The Society for Neuroscience, poster.

6. **Nicholas MA**, Yttri EA. (2024). *Revealing the corticostriatal dynamics underlying trial and error motor learning by recording concurrently across multiple identified cell types*. Gordon Research Conference, poster.

7. **Nicholas MA**, Yttri EA. (2023). *Concurrent corticostriatal dynamics across multiple cell types underlying trial and error learning.* The Society for Neuroscience, poster.

8. Panat RP, Ritchie SM, **Nicholas MA**, Gordon HL, Yuan B, Hu C, Jahan S, Bezbaruah R, Reddy JW, Chamanzar M, Yttri EA. (2022). *3D Printed Customizable Neural Probes.* 12th International Conference on Microelectrode Arrays for Life Sciences 2022, Poster.

9. **Nicholas MA**, Yttri EA. (2021). *Motor cortex but not parietal lesions severely impair reaching, spontaneous motor decisions and striatal dynamics*. Swedish Basal Ganglia Society, poster.

10. **Nicholas MA**, Yttri EA. (2021). *Motor cortex but not parietal lesions severely impair reaching, spontaneous motor decisions and striatal dynamics*. The Society for Neuroscience, poster.

11. **Nicholas MA**. (2020). *Cortical contributions to movement and striatal activity.* Great Hall Of Brain Science Seminar. Carnegie Mellon, talk.

12. **Nicholas MA**, Belsey P, Yttri EA. (2019). *Causal dissection of cortical-striatal interactions governing the neural circuit control of reaching.* The Society for Neuroscience, poster.

13. Hodge AT, **Nicholas MA**, Dudman JT, Yttri EA. (2019). *Striatal stimulation reinforces, but does not select, naturalistic untrained behavior.* The Society for Neuroscience, poster.

14. **Nicholas MA**, Belsey P, Yttri EA. (2019). *Causal dissection of cortical-striatal interactions governing the neural circuit control of reaching.* CMU Department of Biological Sciences Elizabeth Jones Retreat 2019, poster.

15. Panat R, Saleh MS, Ritchie S, **Nicholas MA**, Bezbaruah R, Yttri EA. (2019). *CMU array: A fully-customizable, ultra-high density invasive electrode for large-scale recording and optical stimulation enabled through nanoparticle 3D printing.* The Society for Neuroscience, talk.

16. **Nicholas MA**. (2019). *Causal dissection of cortical-striatal interactions governing the neural circuit control of reaching during goal directed behaviors*. Great Hall Of Brain Science Seminar. Carnegie Mellon, talk.

17. Saleh MS, Ritchie S, **Nicholas MA**, Bezbaruah R, Yttri EA, Panat R. (2019). *CMU Array: A Customizable Ultra-High-Density Optic-Fiber Paired Neural Interface by Nanoparticle 3D Printing.* Biomedical Engineering Society 2019 Annual Meeting, poster.

18. Saleh MS, Bezbaruah R, **Nicholas MA**, Reddy J, Chamanzar M, Yttri EA, Panat R. (2019). *Customizable Ultra-High-Density Optic-Fiber Paired Neural Interfaces by Nanoparticle 3D Printing.* 2019 9th International IEEE/EMBS Conference on Neural Engineering (NER), poster.

19. Saleh MS, **Nicholas MA**, Bezbaruah R, Reddy J, Chamanzar M, Yttri EA, Panat R. (2018). *3D Printed Ultra-High Density, High Aspect Ratio Microelectrode Arrays for Next-Generation Neural Probes and Drug Delivery Applications*. 2018 Materials Research Society, poster.

20. Saleh MS, **Nicholas MA**, Bezbaruah R, Yttri EA, Panat, R. (2018). *Microscale 3D Printing by Additive Nanoparticle Assembly for Energy Storage Systems and Biomedical Devices.* International Mechanical Engineering Congress & Exposition, poster.

21. **Nicholas MA**, Saleh MS, Hodge, AT, Panat R, Yttri EA. (2018). *Customizable, 3D-printed, thousand-channel probes for neural recording and photostimulation.* Cell-NERF Symposium, Neurotechnologies, talk.

22. Kaskan PM, Dean MA, **Nicholas MA**, Mitz AR, Ungerleider LG, Murray EA. (2016) *Gustatory responses in macaque monkeys revealed with fMRI.* The Society for Neuroscience, poster.

23. **Nicholas MA**, Dean MA, Hwang J, Kaskan PM, Murray EA. (2016) *The role of the amygdala in attending to reward predictive visual images.* 18th Annual NIMH DIRP Fellows' Scientific Training Day, poster.

24. Kaskan PM, Dean MA, **Nicholas MA**, Mitz AR, Murray EA. (2015) *Gustatory responses in macaque monkeys revealed with fMRI*. 17th Annual NIMH DIRP Fellows' Scientific Training Day, poster.

25. **Nicholas MA**, Pafundo DE, Kuhlman SJ. (2015) *Corticocortical feedback-mediated facilitation is cell-type specific and gated by nucleus basalis activity.* Carnegie Mellon Meeting of the Minds Poster Presentation, poster.

26. **Nicholas MA**, Pafundo DE, Kuhlman SJ. (2014) *Top-Down Control of Visual Response in Mouse.* Howard Hughes Medical Institute Seminar Presentation, talk.

TECHNICAL SKILLS

Computer: MATLAB, Python, C++, Java, MonkeyLogic, Adobe Illustrator, Adobe Photoshop, ImageJ

Wet Lab: Chronic and acute electrophysiology, Histology, Animal care of rodent and non-human primates, Rodent husbandry

Surgical: Intracranial Implant, Chronic Electrode Implant, Transcardial perfusion, Intracranial injections, Intracranial window implant

PROFESSIONAL MEMBERSHIPS

Society for the Neural Control of Movement — 2020 - Present

Swedish Basal Ganglia Society — 2019 - Present

Society for Neuroscience — 2016 - Present

TEACHING AND MENTORING

In my time working in the Yttri Lab I have served as a mentor to both graduate and undergraduate students. As a graduate student I have had 8 students report directly to me working on both shared and independent projects.

Special Lecturer, Summer Academy for Math and Science, Carnegie Mellon — Summer 2025

Taught lessons on electrophysiology and systems neuroscience through primary literature to roughly 15 highs school students

Special Lecturer, International Youth Neuroscience Association — Summer 2024

Taught material pertaining to learning and memory through primary literature to over 50 highs school and undergraduate students from around the world

Advance Systems Neuroscience Instructor, Carnegie Mellon — Spring 2022 - 2024

Taught graduate students systems neuroscience through primary literature with an additional focus on critical reasoning, discussion and presentation and communication of scientific content, roughly 15 students per semester

2023 course evaluation comment: "Mark is so patient and helpful and passionate. I love the discussions during the classes. And the feedback of presentation is always so detailed and encouraging. Love everything about this class."

2024 course evaluation comment: "Mark always give us good feedback after our presentations, I really appreciate that. He provides us with a very good discussion environment, which I have never experienced before. This discussion makes me have more understanding in each paper, and let me think as a researcher."

Modern Biology Teaching Assistant, Carnegie Mellon — Fall 2020

Assisted during lectures, held weekly Office Hours, created course content and graded material, roughly 150 students

Experimental Neuroscience Teaching Assistant, Carnegie Mellon — Spring 2015

Taught undergraduate students both neuroscience methodology and content in a hands on laboratory environment. Taught some lectures, created course content and graded material, roughly 20 students

OUTREACH AND VOLUNTEERING

Cortico-Basal Ganglia Journal Club, creator, organizer — 2018 - Present

Organize and lead a weekly journal club about motor control, learning and decision making and related topics with members around the Pittsburgh neuroscience community. Along with traditional journal club presentations, I've organized an "In Depth Mini Series" where we discuss 3-4 papers a week for multiple consecutive weeks to explore a given topic in the field. With our transition to virtual presentations, we have also invited external speakers from around the world to share their work as well.

Letters To a Pre-Scientist — 2024 - Present

Program designed to pair scientists with school children to introduce them to the fields of science, technology, engineering and math. Paired with a 6th grade student, we have been corresponding about our shared interests and what life is like studying the brain

During my first year in the program, I received an award for writing exceptionally engaging letters. Comments from the award:

From Student: "Thank you for making every letter really interesting. It made me want to be a neuroscientist. It also made me realize that research and science is really cool."

From Teacher: "[the student] loved reading your letters and writing back to you…Thank you for making this year special for him and for igniting his love of science."

What can we learn from putting electrodes in all parts of the brain? - Science Scavenger Hunt 2022

Outreach event to teach undergraduate students about research being done in the Department of Biological Science, roughly 40 students

A unifying hypothesis behind skilled motor actions — 2022

Interviewed by the Carnegie Mellon newspaper, The Tartan, following a departmental talk to share my findings studying how cortex and striatum work together to produce skilled motor actions

How we use optogenetics to study how the brain controls behavior? - Science Scavenger Hunt 2021

Outreach event to teach undergraduate students about research being done in the Department of Biological Science, roughly 80 students

Colloquium Committee, Center for the Neural Basis of Cognition 2021 - 2023

Organize invited speakers to present to the Pittsburgh neuroscience community

How do neurons talk to each other and how do we listen? - Science Scavenger Hunt 2019

Outreach event to teach undergraduate students about research being done in the Department of Biological Science, roughly 60 students

Department of Biological Science Invited Alumni Career Panel 2019

Shared how my undergraduate education has helped me in academic research with roughly 100 students

GFP in the brain: What is a neuron - Science Scavenger Hunt 2014

Outreach event to teach high school and undergraduate students about histology, microscopy and single neuron morphology, roughly 50 students

Mellon College of Science Mentor 2013 - 2017

Mentored a group of 8 undergraduate students starting before their first year and through their time at the university. I held weekly group meeting during their first semester to discuss available resources and then met individually at least once a semester following to help in their progress.